



Name: _____

Class: _____

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Programming 2 — Answers

The three constructs, and the self-checkout program — OCR J277 §2.2.1

Name the construct — answers

What the program does	Construct	How you can tell
Asks your name, then your age, then says hello	Sequence	Three steps, in order, each once. No question, no repeat.
Keeps asking for a password until you get it right	Iteration	The asking happens <i>again</i> . Repeating is the giveaway.
Prints “Well done” only if your score is over 50	Selection	“Only if” — one path runs, the other does not.
Prints the 5 times table, from 5 up to 60	Iteration	Twelve times through the same instruction.
Charges half price if under 16, full price otherwise	Selection	Two prices, one customer. Exactly one applies.
Turns the heating on, waits an hour, turns it off	Sequence	Three steps in a fixed order. The wait is a step, not a loop.

If you put “Selection” for the password one

That is the mistake almost everyone makes, and it is worth understanding rather than just correcting.

You were not wrong that there is a *question* in it — there is. But **selection asks the question once and then moves on**. This one asks, and if you get it wrong, **asks again**. Something happens more than once, so it is iteration.

The test is not “is there a condition?” Both constructs have conditions. The test is **“does anything happen more than once?”**

On a flowchart it is even easier: **a diamond on its own is selection; a diamond with an arrow going back up the page is iteration.**

The self-checkout — one solution

As with Programming 1, **yours may look different and still be right**. Check what it does, not how it is spelled.

```
items = int(input("How many items? "))
total = 0

for i in range(items):
    price = float(input("Price of item: "))
    total = total + price

print("Total: " + str(total))

if total > 50:
    print("Free delivery!")
else:
    print("Delivery charge: 4.99")
```

All three constructs are in there. Sequence — the lines run top to bottom. Iteration — the for loop, once per item. Selection — the if/else at the end.

Running it with three items priced 20.00, 19.50 and 15.25:

```
Total: 54.75
Free delivery!
```

The bug almost everyone hits — and it gives no error

Put `total = 0` *inside* the loop instead of before it:

```
for i in range(items):
    total = 0          # <-- wrong place
    price = float(input("Price of item: "))
    total = total + price
```

Same three items. Now it prints:

```
Total: 15.25
```

That is just the last item's price. The total is wiped clean every time round the loop, so it never accumulates.

No error message. No red text. It just quietly gives the wrong answer — which makes it a **logic error**, the third kind, and the hardest to find.

Why it happens: anything *inside* a loop runs *every time round*. `total = 0` is a setup step, so it belongs **before** the loop starts — you only want to do it once.

“Where is the selection?” — the trick question

On the worksheet you were asked to find all three constructs in your Programming 1 greeting program:

```
name = input("What is your name? ")
times = int(input("How many times? "))
for i in range(times):
    print("Hello, " + name)
```

The answer: there isn't one

Sequence — lines 1 and 2, running in order. **Iteration** — line 3, the for loop.

Selection — **there is none**. There is no if anywhere, and no question being asked about a value. **The program does not need one**.

That is the whole point of the question. **A program does not have to use all three constructs** — it uses the ones the job requires. Plenty of perfectly good programs never make a decision at all.

If you hunted for a selection and thought you had found one, look again at what `range()` does: it decides *how many times*, not *whether*. Counting is not choosing.

Adding the missing construct

Want the greeting program to use all three? Give it a decision to make:

```
name = input("What is your name? ")
times = int(input("How many times? "))

if times > 10:
    print("That is a lot of greetings!")

for i in range(times):
    print("Hello, " + name)
```

Now it has all three — and notice that the if did not replace anything. It was simply added, because the program needed to decide something it did not need to decide before.